

10.6 The impacts of energy production

- ~~1~~ Why is energy conservation important?
- ~~2~~ What is one way that individuals can save energy at home?
- ~~3~~ Give **one** positive impact of a secure energy supply. [1]
- ~~4~~ Give **two** negative environmental impacts of renewable energy sources. [2]
- ~~5~~ Justify the use of non-renewable energy sources. [3]

REFLECTION

Having now completed a number of questions using the command word 'justify', what do you think makes the response below a good answer to the question? How will you approach 'justify' questions going forwards?

Justify how hydroponics can be a sustainable solution for food production in areas with poor soil. [5]

Hydroponics is sustainable because it allows plants to grow without soil. This is done through the use of nutrient-rich water, making it ideal for areas with poor soil quality such as deserts. It uses up to 90% less water than traditional farming making it effective in water-scarce regions. Hydroponics can be done in urban or indoor spaces.

This reduces the need for farmland and prevents deforestation. It also lowers the use of chemical pesticides and fertilisers reducing the impact on the environment. Hydroponics supports local food production,

therefore cutting the carbon emissions linked to food transport and improving food security.

SELF-ASSESSMENT CHECKLIST

Let's revisit the knowledge focus and exam skills focus for this topic.

Decide how confident you are with each statement.

Now I can	Show it	Needs more work	Almost there	Confident to move on
Describe how food is produced	Describe how aeroponics can be used to grow food.			
Outline global patterns of food supply and demand	Compare food consumption in HICs and LICs.			
Explain the challenges of food supply	Explain two human and natural factors that affect food supply.			
Describe how energy is produced	Identify three renewable energy sources and explain a benefit of each.			
Outline global patterns of	What is meant by the term			

Now I can	Show it	Needs more work	Almost there	Confident to move on
energy supply and demand	energy security?			
Evaluate the impacts and sustainable management of energy production	Evaluate the environmental impacts of burning fossil fuels to generate energy.			
Understand the command word 'give'	Answer a 'give' question confidently, such as 'Give one input used on an arable farm.'			
Understand the command word 'justify'	Answer a 'justify' question confidently, such as 'Justify the use of subsistence farming when there is a growing global demand for food.'			